

Urban Ride Time Prediction

Data Mining Project

HW1

KNTU
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Course: Data Mining

Project Overview

This project focuses on predicting the duration of urban ride trips using data mining and machine learning techniques. The main goal is to analyze trip information and build a predictive system that can estimate how long a ride will take.

Ride duration is affected by several factors, including starting time, ending time, number of riders, trip distance, location, and traffic-related patterns. By studying these features, the project aims to understand which factors have the strongest effect on travel time.

Dataset Description

The dataset contains ride records, where each row represents one completed trip. The main target column is `elapsed_seconds`, which shows the total duration of the trip in seconds and is the value that should be predicted.

The dataset includes a `record_key` column for identifying each trip and a `carrier_code` column that represents the service provider. Time information is included through `start_time` and `end_time`, which show when the trip started and ended.

The dataset also contains `rider_total`, which shows the number of riders in the vehicle. Location information is provided using starting and ending coordinates, including `start_xcoord`, `start_ycoord`, `end_xcoord`, and `end_ycoord`.

Another column, `save_forward_marker`, shows whether the trip record was stored before being sent to the system. These features together describe the main conditions of each ride and can be used to predict trip duration.

Data Cleaning

Ride datasets may contain missing values, duplicated records, invalid coordinates, or unrealistic trip durations. These issues should be handled before building the prediction model.

Important cleaning tasks include checking for missing values, removing duplicate trips, detecting abnormal rider totals, and identifying trips with extremely short or extremely long durations. The coordinate columns `start_xcoord`, `start_ycoord`, `end_xcoord`, and `end_ycoord` should also be checked to make sure starting and ending locations are valid.

Some trips may have unrealistic distances or travel speeds. These outliers should be

handled carefully because they can strongly affect the accuracy of the model.

Feature Engineering

Feature engineering is an important part of this project because the original columns can be transformed into more useful information.

The `start_time` column can be used to create time-based features such as start hour, day of week, month, weekend indicator, and rush-hour indicator. These features help capture traffic and travel patterns during different times.

The starting and ending coordinates can be used to estimate the distance between two locations. Distance is one of the most important features for predicting trip duration because longer trips usually take more time.

The `save_forward_marker` and `carrier_code` columns can also be prepared as categorical features so they can be used by machine learning algorithms.

Modeling and Evaluation

After preprocessing the data, suitable regression models can be trained to predict ride duration. Different approaches may be compared to find the model that performs best on unseen data.

The models can be evaluated using regression metrics such as RMSE or MAE. The final result should estimate ride duration and provide useful insight into the factors that influence travel time.